

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY****Third Semester of B.Tech. (CL) Examination****March-Apr 2018****CL242 Surveying****Date: 30.03.2018, Friday****Time: 01:30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

**SECTION – I****Q - 1 Answer the question below.****[07]**

- (a) The operation of revolving the telescope through  $180^\circ$  in the vertical plane is known as \_\_\_\_\_
- (a) Swinging (b) Inverted Telescope  
(c) Transiting (d) Face Right
- (b) When the length of any chord of a curve is less than peg interval, it is known as a
- (b) Small chord (b) Sub-chord  
(c) Normal chord (d) Short chord
- (c) Define Invert.
- (d) What is the purpose of Setting out works?
- (e) Which of the following is associated with the setting out of sewers and pipe lines?
- (a) Batter-board (b) Cross-staff  
(c) Cross-head (d) Optical square
- (f) In EDM “set screw” is used to adjust the \_\_\_\_\_
- (a) Clarity of thread (b) Clarity of lens  
(c) Clarity of object (d) Clarity of vision
- (g) Which of the following instruments is enabled with Autofocus function?
- (a) Electronic Distance Measurement (b) Total Station  
(c) Electronic theodolite (d) All of the above

- Q – 2 (a) i. Classify the various types of curve with its definitions. [04]**
- ii. Describe the operations involved while measuring distance with EDM. [03]**
- (b) Areas within the contour line at the site of reservoir and the face of the proposed dam [04]**  
are as follows:

|                                 |         |         |         |         |        |
|---------------------------------|---------|---------|---------|---------|--------|
| <b>Contour<br/>(m)</b>          | 101     | 102     | 103     | 104     | 105    |
| <b>Area<br/>(m<sup>2</sup>)</b> | 1000    | 12800   | 95200   | 147600  | 872500 |
| <b>Contour<br/>(m)</b>          | 106     | 107     | 108     | 109     | ---    |
| <b>Area<br/>(m<sup>2</sup>)</b> | 1350000 | 1985000 | 2286000 | 2512000 | ---    |

Determine the capacity (volume) of the reservoir by trapezoidal and prismoidal formula.

- (c) Explain the procedure of Setting out grade stake for sewer. [03]

**OR**

- (b) Calculate the area of a plan from the following readings of a planimeter. [04]

Initial reading = 8.348

Final reading = 1.435

The zero of the disc passed the fixed index mark twice in the clockwise direction.

The anchor point was placed outside the plan and the tracing point was moved in the clockwise direction. Take  $M = 100\text{cm}^2$ ,  $C = 0$ .

- (c) Explain the commencing task before setting out works. [03]

- Q – 3** (a) Two tangents intersect at chainage of 4242 m. Angle of intersection is  $140^\circ$ . [07]

Calculate all necessary data to set out the curve of radius 344 m by Offsets from the chord produced method. Peg interval and chain length is 30 m.

- (b) Discuss and derive the following methods of computation of area with straight but irregular boundaries. [07]

(a) Mid-ordinate rule      (b) Average-ordinate rule      (c) Trapezoidal rule

**OR**

- Q – 3** (a) Define the following [07]

Forward Tangent, Point of Curvature, Tangent distance, Deflection angle, Angle of Intersection, Long chord and Mid-ordinate.

- (b) A railway embankment is 12m wide with side slope 1.5 to 1. Assuming the ground to [07]

be level in a direction transverse to the center line. Calculate the volume by trapezoidal and prismoidal formula, contained in a length of 132 m, the center height at 22m interval being in a meters 2.3, 3.8, 4.0, 4.2, 3.7, 2.8, 2.4.

## SECTION – II

**Q - 4     Answer the question below. [07]**

- (a) Vertical distance between two successive contour lines is called as \_\_\_\_\_. (Ridge / Contour Interval / Valley)
- (b) In ocean where depth of water is great \_\_\_\_\_ instrument is used. ( lead line / Fathometer / Sounding pole)
- (c) For leveling plane table, \_\_\_\_\_ is used. (U-fork / Tripod / Spirit Level)
- (d) Define Plunging.
- (e) Inaccessible points may be located by \_\_\_\_\_ method. (Radiation / Intersection / Resection)
- (f) In closed traverse, algebraic sum of the latitudes and departures must be equal to \_\_\_\_\_. ( $90^\circ$  /  $180^\circ$  /  $0^\circ$ )
- (g) When vertical circle is to the right side of the observer, it is called telescope \_\_\_\_\_. (Inverted, Reversed, Normal)

**Q – 5     (a)     Enlist the accessory used in Plane table surveying? Explain with sketch Alidade and U-fork accessory in detail. [05]**

**OR**

- (a) Explain in details various errors in plane table surveying. [05]
- (b) What is sounding? Explain various instruments used in sounding and explain any one in detail. [05]
- (c) Define the term contour. Explain the uses of contour map in the field of Civil Engineering. [04]

**OR**

- (c) Show a neat sketches the characteristics features of contour lines of the following [04]
  - 1. Pond                      2. Hill                      3. A Valley Line                      4. Ridge Line

**Q – 6     (a)     List out the methods of taking Horizontal angles with Theodolite and Explain any one method of taking horizontal angles with vernier transit Theodolite. [05]**

- (b) Prepare Gale's Traverse table to adjust the closing error of the closed traverse ABCDA for Following data [07]

| Line (L)         | AB   | BC   | CD   | DA   |
|------------------|------|------|------|------|
| Length (m)       | 110  | 80   | 95   | 160  |
| Corrected W.C.B. | 110° | 170° | 250° | 350° |

- (c) Define the following terms. (Attempt any two) [02]  
 1. Face Left observation      2. Swinging      3. Axis of Telescope

**OR**

- Q – 6 (a) List the various methods of traversing? Explain fast needle method in detail. [05]
- (b) Define the following terms and also write equation of each [07]  
 (1) Latitude      (2) Departure      (3) Closing error      (4) Balancing of traverse
- (c) What is meant by theodolite? Explain various types of theodolite. [02]

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